

The BERT Walked Down the Garden Path Assigned Semantic Roles

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INTRODUCTION

The horse **raced** past the barn fell.

- Garden path sentences (like the one above) are difficult to process because **they are temporarily ambiguous**
- Raced* is probably assumed to be the action done by the horse until we reach *fell...* which has no agent! Somehow this sentence will have to be reanalyzed to reach the correct interpretation
- The difficulties in garden path processing arise because of the **sequential nature of human language processing**
- Garden path sentences cause **difficulties for both syntactic parsing and semantic role assignment**

QUESTION ANSWERING

40x4 Sentences [3]

5x Question Types

Garden Path	The child _{1,3} (who was) bought ₅ the ice cream cone hugged ₄ the nanny ₂
Local Coherence	The nanny ₁ hugged ₄ the child _{2,3} (who was) bought ₅ the ice cream cone

¹Agent Matrix	Q: Who gave a hug? A: {The child/the nanny}
²Patient Matrix	Q: Who got a hug? A: {The nanny/the child}
³Ambiguous Arg.	Q: Who got an ice cream cone? A: The child
⁴Matrix Action	Q: What did {the child/the nanny} do? A: Hugged
⁵Embedded Act.	Q: What did the child get? A: Bought the ice cream cone

Behavioral Experiment

- 74 English speakers recruited from Prolific
- 45 random items: 32 target + 13 filler
- > 80 responses for each of the sentence and question types

The nanny hugged the child
bought the ice cream cone.

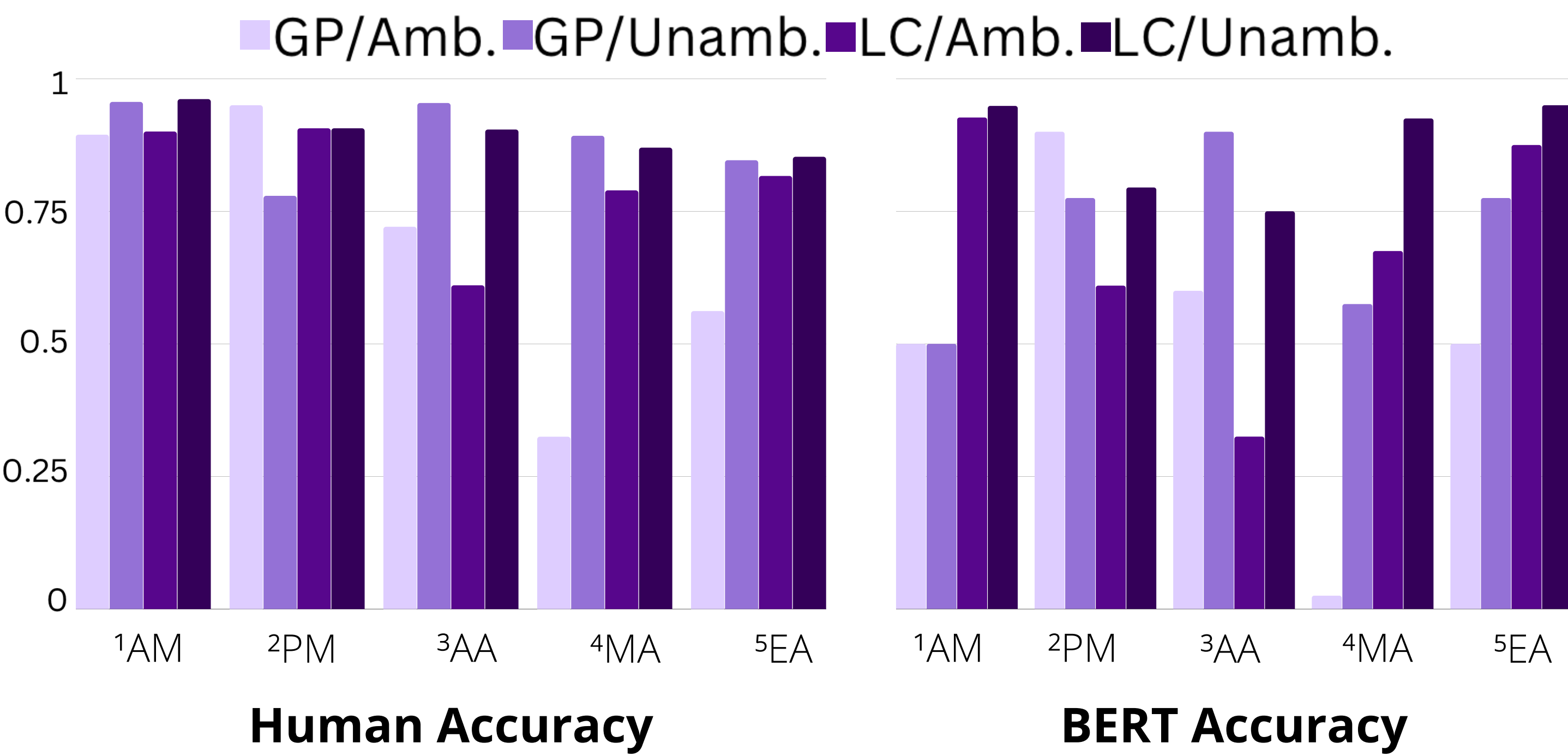
Who hugged the child?

Context: The nanny hugged the
child bought the ice cream cone.
Question: Who hugged the
child?



BERT Experiment

- Same 800 context + question pairs
- Pre-trained BERT_{BASE} model which had been fine-tuned for QA using SQuAD2.0



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OBJECTIVES

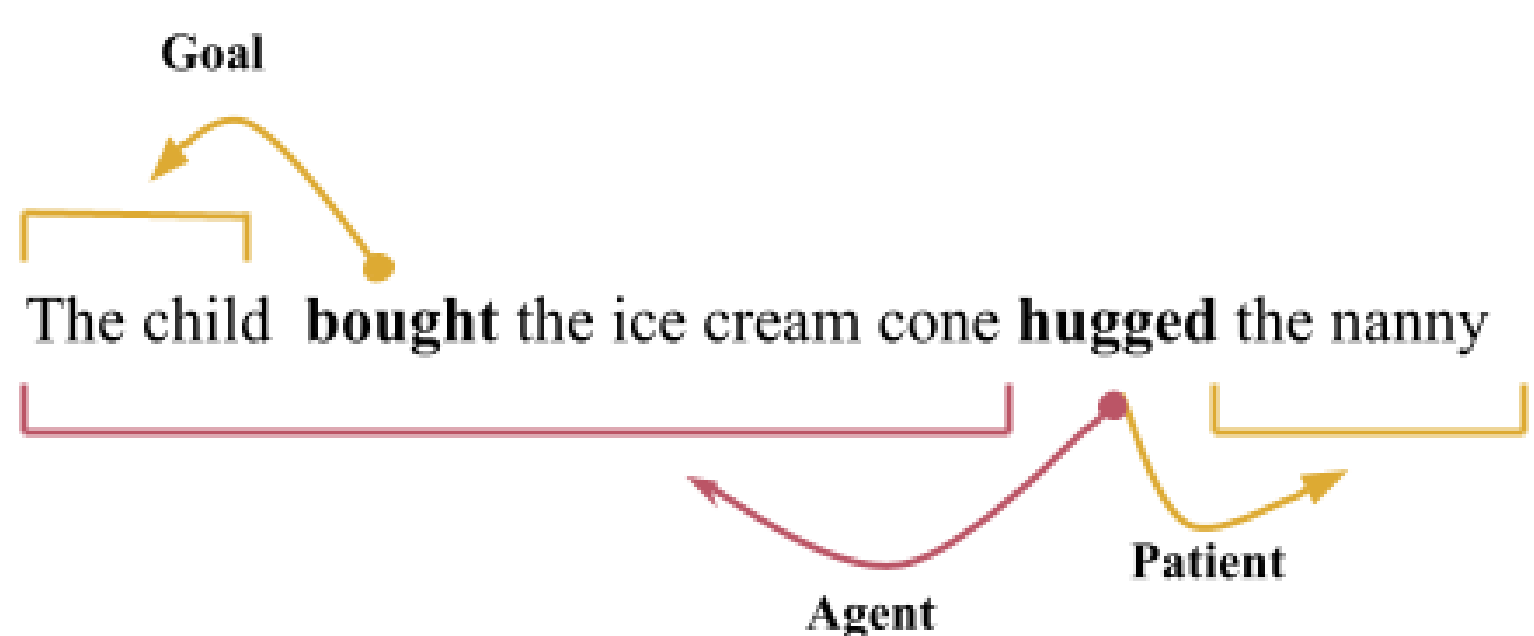
Question 1: Given that BERT and humans process language differently, do the two also differ in the extent to which they can extract information from garden path sentences in order to correctly answer questions about them?

Question 2: Humans might have difficulty processing garden path sentences because of semantic role misassignment. Is this also a challenge that BERT faces when encountering garden paths?

PROBE TASK

Probe Training

We trained two binary linear classifiers on BERT's word embeddings (8th layer, following findings from [6]) from sentences annotated in PropBank [5] for semantic roles Agent and Goal.

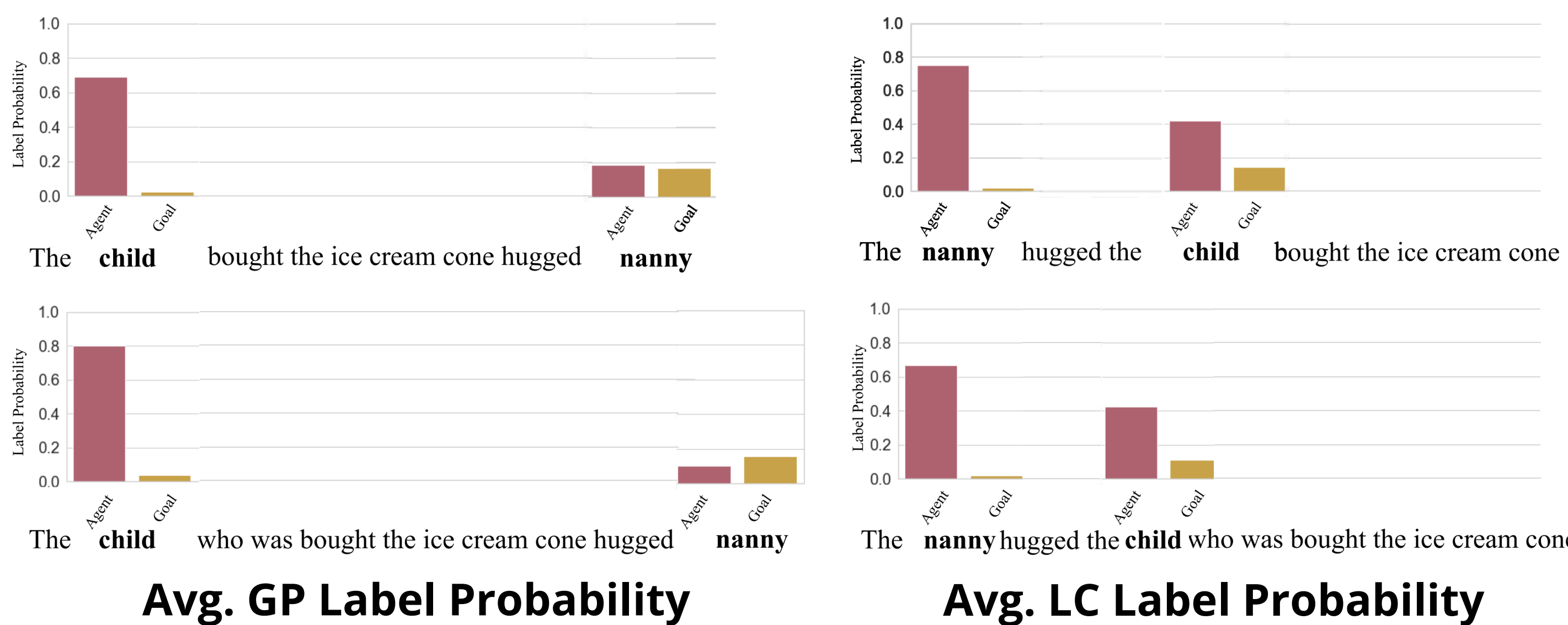


Classifier	Accuracy	Selectivity
Agent Classifier	85	35
Goal Classifier	85	37

- 5000 training examples with a 50/50 split between having or not having the role in question.
- Chose hyperparameters to maximize selectivity [6]
- 80/20 train test split
- High accuracy on Propbank test sentences

Probe Testing on Garden Paths

- Tested on subset of context sentences from the question answering section, selecting only sentences containing both Agent and Goal roles, according to PropBank annotations (97/160 sentences total)



- BERT fails to assign the Goal role to the head of the embedded clause
- Nouns immediately preceding verbs are most likely to be agents regardless of the syntactic parse of the structure

CONCLUSION

- BERT is worse than humans at answering questions about garden path sentences** despite not being subject to the ambiguity
- It also shows difficulty in some cases with no ambiguity, suggesting **difficulty processing relative clauses** as well
- These results may be due to a **failure to assign the correct semantic role** in relative clauses, following theories from human sentence processing

Poster PDF
and Code



tinyurl.com/emnlpBERT

